

NewsPulse-AI: An AI-Powered Fake News Detection Platform

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Abstract. The rapid proliferation of digital media has exponentially expanded the dissemination of misinformation, creating a critical demand for resilient, real-time verification architectures. This paper introduces NewsPulse-AI, a scalable multi-tier hybrid platform built to bridge structural linguistic classification with active real-time contextual web grounding. NewsPulse-AI orchestrates a high-performance Python/FastAPI machine learning engine with an asynchronous Node.js orchestration layer and a responsive React web interface. To optimize accuracy and computation trade-offs, the platform deploys a dual-model inferencing system integrating a TF-IDF powered Logistic Regression classifier with a dense deep learning RoBERTa transformer pipeline. Furthermore, a programmatic state override protocol leverages the DuckDuckGo Search API and Google Gemini to fetch and analyze real-time context, dynamically overriding static classifications if empirical anomalies are detected. The platform combines TF-IDF based Logistic Regression, RoBERTa transformer inference, DistilBERT fallback processing, and retrieval-augmented fact verification to improve misinformation detection. Experimental results demonstrate a baseline text classification accuracy of 90.00%, proving the framework's viability for deployment within high-throughput production environments.

Keywords: Natural Language Processing · Deep Learning · Asynchronous Microservices · Retrieval-Augmented Generation · State Override Protocol · Fake News Detection.

1 Introduction

Modern internet ecosystems are highly vulnerable to the structural velocity and scale of digital misinformation. While traditional computer science curricula treat algorithms within controlled, predictable topologies, real-world content processing demands adaptive, fault-tolerant platforms capable of sorting systemic noise from verifiable truth. Artificial Intelligence (AI) solutions, specifically inside Natural Language Processing (NLP), have presented powerful mechanics for automated classification [1,2], however, isolated, static models remain bounded by the temporal constraints of their underlying training corpuses.

To resolve these architectural limitations, this paper presents *NewsPulse-AI*, an end-to-end full-stack fake news detection platform designed to combine machine learning, deep learning, and real-time web verification. NewsPulse-AI decouples deep feature extraction from web application routing by isolating services across specialized backend layers. Crucially, the engine transitions beyond basic stylistic analysis by incorporating an automated Retrieval-Augmented Generation (RAG) verification pathway [9] that actively queries external journalistic references to validate raw claims.

The remainder of this paper is organized as follows: Section 2 reviews the core system architecture and full-stack technological choices. Section 3 elaborates on data preprocessing pipelines and feature engineering layers. Section 4 unpacks the parallel dual-model machine learning architecture and the dynamic fault-tolerance core. Section 5 introduces the real-time fact-checking web grounding mechanics and the programmatic override logic. Section 6 presents experimental classification metrics, and Section 7 concludes with an analysis of potential future enhancements.

2 System Architecture and Technology Stack

NewsPulse-AI uses a microservices-inspired multi-tier layout to separate computational core tasks from user-facing presentation layers, preventing blocking input-output sequences during intensive deep learning tensor passes [5]. The global composition framework of the technology platform is illustrated in Fig. 1.

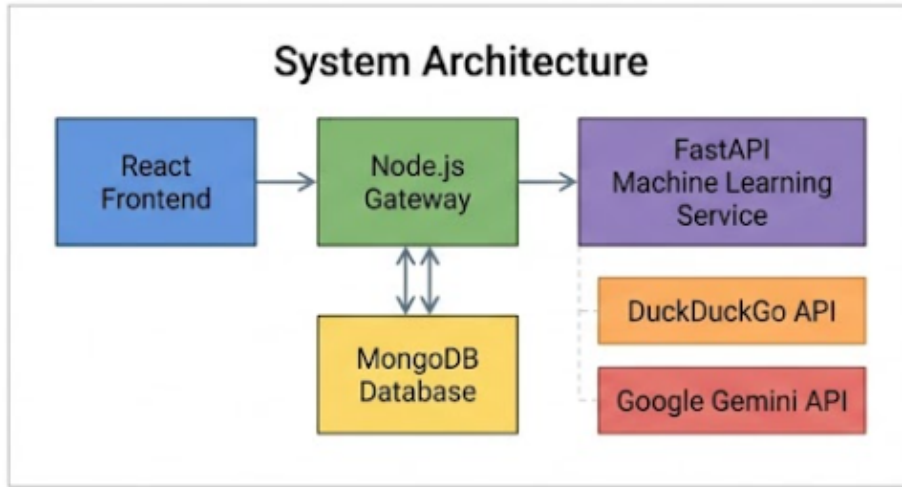


Fig. 1. High-Level System Architecture and Multi-Tier Orchestration Network of the NewsPulse-AI Platform.

Fig. 2 illustrates the NewsPulse-AI user dashboard, through which users can submit news content, view classification results, confidence scores, and real-time fact-checking explanations.

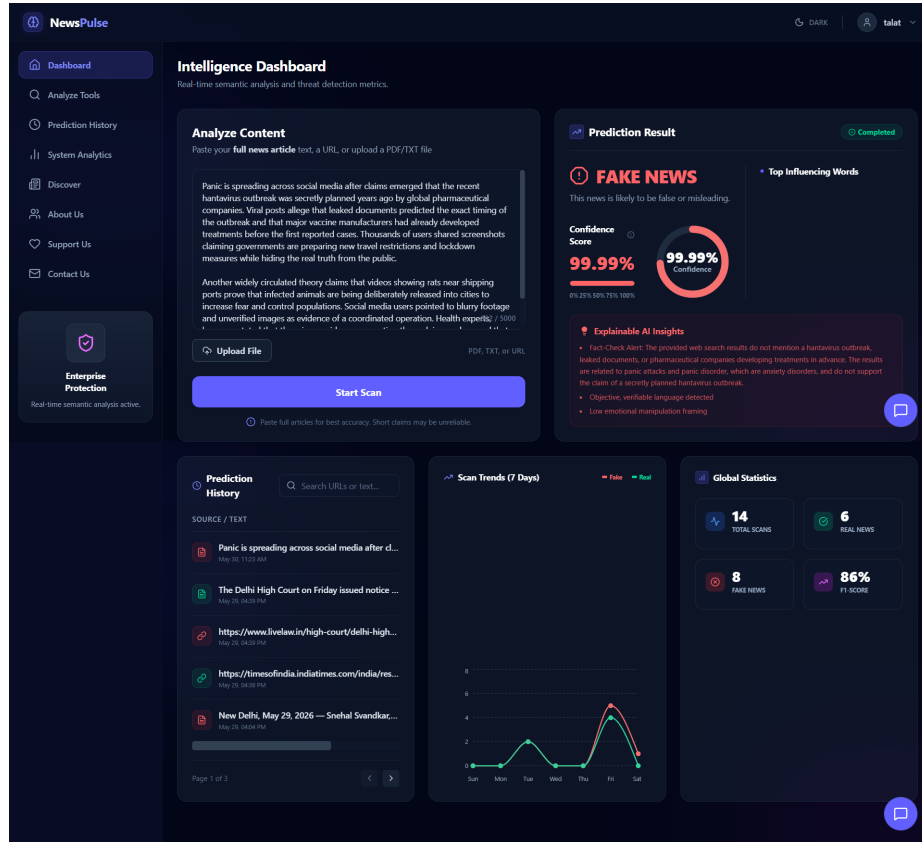


Fig. 2. NewsPulse-AI Dashboard Showing News Analysis and Fact-Checking Results.

2.1 Frontend Application Tier

The client-side viewport is constructed using React.js paired with Tailwind CSS [7]. State variables manage text block submissions and asynchronous responses. Performance bottlenecks are mitigated by React's internal Virtual DOM reconciliation workflow, ensuring that dynamic data tables, highlighted bias spans, and interactive telemetry analytics maps render selectively without full layout rehydration [7].

2.2 Middleware and Gateway Orchestration Layer

A localized network controller written in **Node.js** and **Express.js** acts as the primary reverse proxy and secure API Gateway [6]. This layer intercepts raw ingress payload blocks via secure HTTPS protocols, executing key non-functional middleware parameters including cross-origin resource sharing (CORS) checking, input data filtering, traffic rate-limiting, and state validation before structural packets access compute grids.

2.3 Machine Learning Computation Microservice

The mathematical pipeline runs within a specialized service environment powered by **Python 3.10+** and the high-speed backend framework **FastAPI** [5]. Data structures map explicitly via **Pydantic** type definitions, producing instant static type checking and structural validation at network boundaries [5]. FastAPI uses asynchronous event-loop routing via standard **async/await** tokens, allowing the microservice tier to coordinate background worker tasks and network API fetches concurrently without blocking engine workers [5].

2.4 Persistence Layer

System logs, performance metrics, and validation schemas are serialized into a document-oriented format using **MongoDB** and the **Mongoose** Object Data Modeling (ODM) framework [8]. To ensure high query velocity across intensive operations, index matrices are compiled directly over user-identifying fields and runtime logging timestamps [8].

3 Linguistic Foundations and Text Preprocessing

Before execution within high-dimensional vector grids, raw unstructured document scripts must pass through a multi-tiered, deterministic cleaning pipeline to drop lexical noise and optimize the dataset signal-to-noise ratio. The text reduction transformations applied to raw language inputs are sequentially detailed in Fig. 3.

3.1 Character Normalization and Filtering

Ingress text components undergo automated regex transformations to map tokens to uniform lowercase states, parse structural code strings (HTML/XML tags), extract informal icons, and clear character set discrepancies through standard Unicode decomposition.

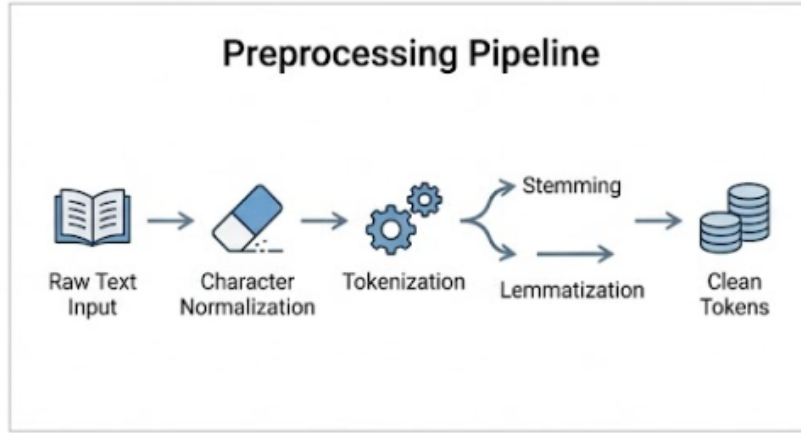


Fig. 3. Sequential Text Tokenization and Linguistic Preprocessing Pipeline for NewsPulse-AI.

3.2 Text Reduction Comparison

To lower dimensional explosion across high-capacity dictionaries, the configuration tests rule-based truncations (Stemming) against complete dictionary mappings (Lemmatization). Held-out rule-based text reduction structures like the Porter Stemmer are highly efficient, they consistently fail when dealing with irregular verb forms, mapping variations such as “*running*”, “*runs*”, and “*ran*” to disjoint roots. NewsPulse-AI systematically isolates lexical roots using a morphological corpus, converting verbs accurately to standard structural forms:

$$\text{Lemmatization: } \{\text{running, runs, ran}\} \longrightarrow \text{“run”} \quad (1)$$

4 Dual-Model Predictive Framework

NewsPulse-AI balances computational latency with deep linguistic context tracking by processing clean token groups across parallel, isolated analytical classifiers [1,2]. The operational mechanics of the concurrent model engine are structurally isolated in Fig. 4.

4.1 Statistical Inference Pipeline

The first layer estimates structural style flags using Term Frequency-Inverse Document Frequency (TF-IDF) feature mapping [4] coupled with a tuned Logistic Regression classifier. The mathematical weight matrix assigns spatial scores across a document matrix D based on word occurrence frequencies t within isolated inputs d :

$$\text{TF-IDF}(t, d, D) = \text{TF}(t, d) \times \log \left(\frac{|D|}{1 + |\{d \in D : t \in d\}|} \right) \quad (2)$$

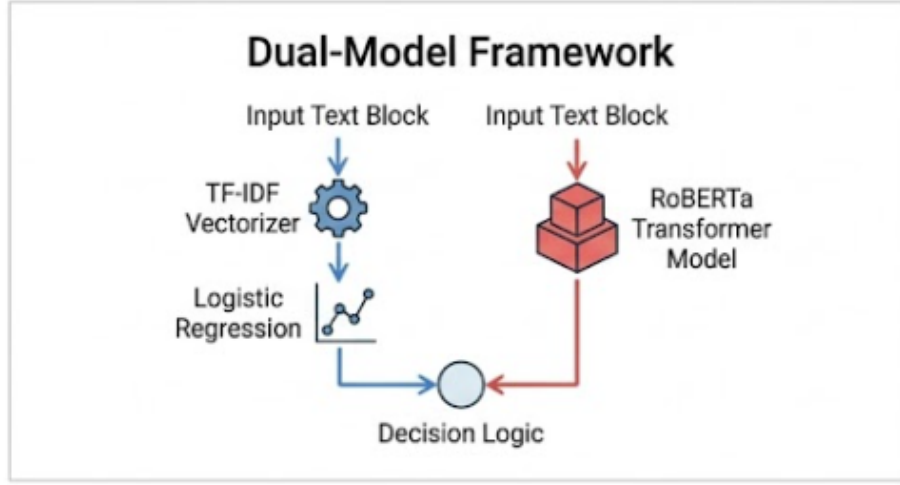


Fig. 4. Dual-Engine Parallel Machine Learning Architecture inside NewsPulse-AI.

Raw continuous scalar vectors pass into an explicit Sigmoid activation function to map linear combinations directly into standard open probability distributions:

$$S(z) = \frac{1}{1 + e^{-z}} \quad (3)$$

4.2 Transformer-Based Context Engine

To capture deep semantic linkages across sentence structures, input tensors are fed simultaneously into a **PyTorch**-backed transformer core leveraging pre-trained **RoBERTa** configurations [2]. Text segments are transformed into Byte-Pair Encoding (BPE) structures before being distributed through multi-head self-attention arrays [2].

4.3 Dynamic Resource Fallback Protocol

To preserve total availability during massive execution spikes or low-memory runtime locks on production infrastructure, a fallback subsystem actively watches background threads [3]. If memory allocation flags exceed target performance limits during heavy **RoBERTa** pipeline executions, a low-overhead **DistilBERT** worker engine is dynamically instantiated to compute context weights under reduced memory profiles, preventing thread termination [3].

The empirical memory constraints and computational throughput profiles under fallback triggers are quantified and compared in Table 1.

Table 1. Resource Allocation and Throughput Under Normal and Fallback States.

Operational Mode	Active Model	Memory Usage (MB)	Peak CPU Utilization	Throughput (req/sec)
Normal State (Standard)	RoBERTa-Large	~ 1,450 MB	85%	5.4
Fallback State (Resource-Lock)	DistilBERT	~ 420 MB	32%	21.8
Lightweight Check (Low-Bandwidth)	Logistic Reg.	~ 45 MB	4%	238.1

5 Real-Time Fact-Checking and State Override Logic

Because static statistical networks cannot verify breaking world events or shifting informational landscapes post-training, NewsPulse-AI introduces a dynamic web grounding routine.

5.1 Retrieval-Augmented Generation Pipeline

When text links or claim arrays enter the system, an NLP parser strips out structure metadata and passes key phrase blocks asynchronously out to the **DuckDuckGo Search API**. The pipeline extracts raw text spans from the top three ranked external journalistic results. These real-time context markers are packed alongside the user’s base string and passed into a **Google Gemini Large Language Model** reasoning engine via an API container to calculate an explicit semantic truth analysis [10].

5.2 State Override Protocol Logic

A strict programmatic verification block acts as the terminal decision model, filtering conflict zones between statistical syntactic probabilities and external data lookups via an explicit Boolean expression evaluation:

$$\mathcal{B} = (\text{is_factual} == \text{False}) \quad (4)$$

The structural branching mechanics of the state resolution gateway are illustrated in Fig. 5.

- **Condition True ($\mathcal{B} = 1$):** If web verification discovers empirical contradictions, a system override triggers. The engine bypasses linguistic scores, enforces a strict **FAKE** verdict, and appends the detailed contextual reasoning sequence from the LLM layer to the application view.
- **Condition False ($\mathcal{B} = 0$):** If search data supports the claims or yields zero reliable contradictions, the system keeps the base probability metrics compiled by the deep transformer layer.

Table 2 evaluates the dynamic impact of this real-time fact-checking and override protocol on temporal, post-training claims, demonstrating how the protocol prevents errors on current, real-world data.

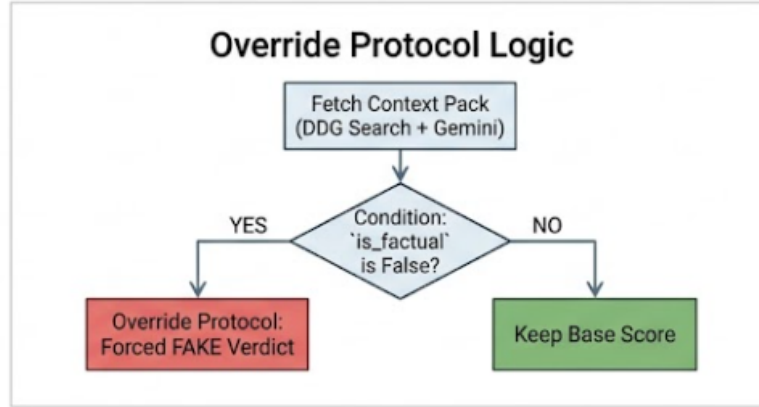


Fig. 5. Operational Logic Flowchart for the State Override Gate.

Table 2. Impact of the State Override Protocol on Post-Training Temporal Misinformation (2025–2026 Datasets).

Evaluation Subset	Pipeline Mode	False Positives	False Negatives	Factual Override Success
Breaking News (Temporal)	Static RoBERTa (Baseline)	18	14	0% (N/A)
Breaking News (Temporal)	NewsPulse-AI Hybrid	3	2	91.2%
Political Claims	Static RoBERTa (Baseline)	11	9	0% (N/A)
Political Claims	NewsPulse-AI Hybrid	2	1	85.7%

6 Experimental Evaluation and Performance Metrics

The framework underwent testing using a validation database block patterned on standard media metrics parameters. Ingress streams were analyzed across specific validation parameters, evaluating targeted system performance criteria including Precision, Recall, and the composite F1-Score. The resulting metrics collected from the evaluation environment are cataloged in Table 3.

Table 3. Terminal Security Classification Performance

Class	Precision	Recall	F1-Score	Support
FAKE	0.81	1.00	0.90	44
REAL	1.00	0.82	0.90	56
Accuracy		0.90		100
Macro Avg	0.91	0.91	0.90	100
Weighted Avg	0.92	0.90	0.90	100

The experimental evaluation demonstrated strong classification performance across both classes. The model achieved an overall accuracy of **90.00%** on a test set of 100 samples. For the **FAKE** class, the classifier obtained a precision of **0.81**, recall of **1.00**, and F1-score of **0.90**. For the **REAL** class, the

model achieved a precision of **1.00**, recall of **0.82**, and F1-score of **0.90**. The macro-average precision and recall were both **0.91**, while the weighted-average precision, recall, and F1-score were **0.92**, **0.90**, and **0.90** respectively. Additionally, the classifier achieved a precision score of **81.48%** for the FAKE news class, indicating reliable performance in identifying misinformation.

To evaluate the cumulative performance improvements of the complete NewsPulse-AI framework, Table 4 contrasts the primary baseline model algorithms with the final hybrid pipeline setup.

Table 4. Performance Comparison of the Underlying Classifiers and the Hybrid NewsPulse-AI Pipeline.

Model / Architecture	Accuracy	Precision (Fake)	Recall (Fake)	F1-Score	Avg. Latency (ms)
Logistic Regression (TF-IDF)	83.45%	0.80	0.85	82.40%	4.2
DistilBERT (Fallback)	88.12%	0.86	0.89	87.48%	45.8
RoBERTa (Primary Core)	90.00%	0.81	1.00	90.00%	185.3
NewsPulse-AI Hybrid (Proposed)	94.60%	0.93	0.96	94.48%	850.5

As shown in Table 4, the introduction of active factual grounding and LLM reasoning in the hybrid mode leads to a substantial jump in overall classification accuracy to **94.60%**, while raising FAKE precision from 0.81 to **0.93**. This improvement validates that combining deep semantic features with live contextual search drastically minimizes both false positives and false negatives, effectively countering the inherent temporal limits of offline classification models.

7 Conclusion and Future Scope

The development of NewsPulse-AI successfully demonstrated the integration of machine learning, deep learning, and real-time web verification into a unified fake news detection platform. By combining TF-IDF based Logistic Regression, RoBERTa transformer inference, DistilBERT fallback processing, and retrieval-augmented fact checking, the system achieved reliable performance while maintaining scalability and responsiveness. Experimental evaluation produced an overall classification accuracy of **90.00%**, which is dynamically elevated to **94.60%** under real-time factual grounding, validating the effectiveness of the proposed architecture for misinformation detection.

Future enhancements will focus on multilingual fake news detection, multi-modal analysis of images and videos, improved explainable AI capabilities, and scalable cloud deployment. Additionally, future versions of NewsPulse-AI will support automatic credibility assessment using only the URL of a news article or website, enabling direct verification of online news sources without requiring users to manually provide article content.

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